FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –2
TEST DATE: 11-12-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. C

Sol. $i = i_0\left(1-e^{-t/\tau}\right)$

$\frac{di}{dt} = \frac{i}{\tau}e^{-t/\tau}$

$U = \frac{1}{2}Li^2$

$\frac{dU}{dt} = \frac{1}{2}L\left(2i\frac{di}{dt}\right) = ke^{-t}(1-e^{-t})$

at t = 0, $\frac{dU}{dt} = 0$

$t \to \infty$, $\frac{du}{dt} = 0$

2. A

Sol. $dC = \frac{k\varepsilon_0 A}{dx}$

$\frac{1}{C_{eV}} = \int\frac{1}{dC} = \int\frac{1}{k\varepsilon_0 A}dx$

$= \int_0^d \frac{1}{k_0 e^{\lambda x}\varepsilon_0 A}dx$

$C_{eq} = \frac{\lambda k_0 \varepsilon_0 A}{1-e^{-\lambda d}}$

3. B

Sol. $H = -kA\dfrac{d\theta}{dx}$

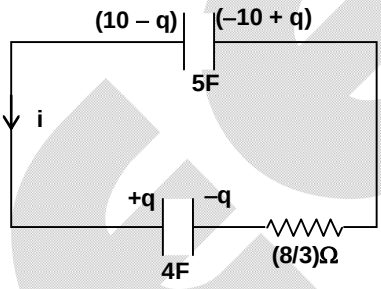
The heat current decreases from the end P and Q through the rod hence slope of (T-x) graph also decreases.

4. C

Sol. $U = \dfrac{kq_0(2q)}{\sqrt{z^2+2a^2}}$

5. A, B, D

Sol. At time 't'



$$-\frac{q}{4} - i\times\frac{8}{3} + \frac{10-q}{5} = 0$$

$$\int_0^q \frac{dq}{2-\frac{9}{20}q} = \int_0^t \frac{3}{8}dt$$

$$q = \frac{40}{9}\left[1 - e^{-\frac{27}{160}t}\right],$$

$$\frac{dq}{dt} = i = \frac{3}{4}e^{-\frac{27}{160}t}$$

$$U_i = \frac{(10)^2}{2\times 5}$$

$$U_f = \frac{(40/9)^2}{2\times 4} + \frac{(50/9)^2}{2\times 5}$$

$$\Delta U = U_i - U_f = \frac{40}{9}J$$

6. B, C

Sol. Current in the circuit

$$i = \frac{5}{R+\frac{R}{2}} = \frac{20}{3R}$$

$\Rightarrow$ Potential drop across PS = $iR = \dfrac{20}{3}V$

Potential drop across SQ = $i\left(\dfrac{R}{2}\right) = \dfrac{10}{3}V$

$\Rightarrow$ Potential q radiant for PS = $\dfrac{20/3}{5} = \dfrac{4}{3}$ V/m

Potential q radiant for SQ = $\dfrac{10/3}{5} = \dfrac{2}{3}$ V/m

$\Rightarrow$ Length required for 5V = $\dfrac{5}{4/3}$ V= 3.75 m

Length required for 8V = $5 + \dfrac{8-\frac{20}{3}}{2/3} = 7$ m

7. A, C

Sol. When charge particle is moved from 'S' to infinity then induced electric field is perpendicular to the displacement hence work done by external agent is zero.

From P to infinity

Work done = q(change in potential difference)

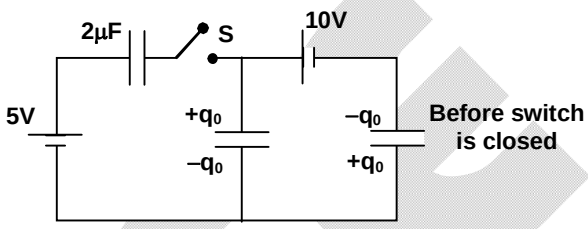
$= q = \frac{\pi r^2}{4}\frac{dB}{dt} = \frac{55}{8}\mu J$

8. A, C, D

Sol. Before switch is closed

$10 - q_0 - \frac{q_0}{2} = 0$

$q_0 = \frac{20}{3}\mu C$



After switch is closed

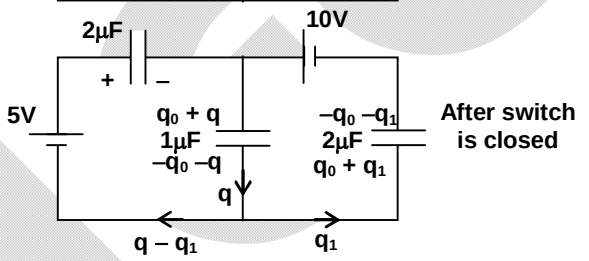
From KVL,

$5 - \frac{q - q_1}{2} - \frac{q_0 + q}{1} = 0$

$10 - \frac{q_0 + q}{1} - \frac{q_0 + q}{2} = 0$

On solving,

$q = -\frac{2}{3}\mu C,\ \ q_1 = \frac{4}{3}\mu C$



$U_i = \frac{1}{2}\frac{(20/3)^2}{1} + \frac{1}{2}\frac{(20/3)^2}{2} = \frac{100}{3}\mu J$

$U_f = \frac{1}{2}\frac{(6)^2}{1} + \frac{1}{2}\frac{(8)^2}{2} + \frac{1}{2}\frac{(2)^2}{2} = 35\ \mu J$

$W_B = 5(-2) + 10 \times \frac{4}{3} = \frac{10}{3}\mu J$

So, heat required = $\frac{5}{3}\mu J$

9. A

Sol. Potential at 'C' = $\frac{kQ}{R} = \frac{\sigma R}{2\varepsilon_0}$

Flat surface of hemispherical shell is on equipotential surface.

10. B, C

Sol. $f_f = \mu mg$

$a = \mu g$

$\int_{v_0}^{v} dv = \int_{0}^{t} -\mu g\, dt$

$v - v_0 = -\mu g t$

$v = v_0 - \mu g t$

$\Rightarrow \quad L\left(-\frac{dm}{dt}\right) = f_f v = \mu mg(v_0 - \mu g t)$

$$\int_{m_0}^{m}\left(-\frac{dm}{m}\right)=\int_{0}^{t}-\frac{\mu g}{L}(v_0-\mu gt)dt$$

$$\ell n\frac{m}{m_0}=-\frac{\mu g}{L}\left(v_0t-\frac{\mu gt^2}{2}\right)$$

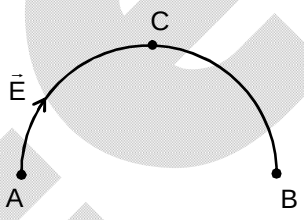
$$m=m_0e^{-\frac{\mu g}{L}\left(v_0t-\frac{\mu gt^2}{2}\right)}$$

Section – B

11. 4

Sol. $\frac{1}{2}mv_C^2=qE\cdot\frac{\pi R}{2}$ …(i)

$\frac{1}{2}mv_B^2=qE\cdot\pi R$ …(ii)



At C, $a_C=\frac{v_C^2}{R}=\frac{qE\pi}{m}$

At B, $a_C=\frac{v_B^2}{R}=\frac{2qE\pi}{m}$

$$a_1=\sqrt{\left(\frac{qE}{m}\right)^2+\left(\frac{qE\pi}{m}\right)^2}$$

$$a_2=\sqrt{\left(\frac{qE}{m}\right)^2+\left(\frac{2qE\pi}{m}\right)^2}$$

$$\frac{a_1}{a_2}=\sqrt{\frac{1+\pi^2}{1+4\pi^2}}$$

12. 2

Sol. $\Delta\ell_1=L_1\alpha_1\Delta T=\frac{30}{100}(1.7\times10^{-5})\times100=0.51\,mm$

$\Delta\ell_2=(1.91-0.51)=1.4$ mm

So, $\Delta\ell_2=L_2\alpha_2\Delta T$

$\alpha_2=2\times10^{-5}/°C$

13. 2

Sol. $\vec{B}=4\hat{i}+\hat{j}+\hat{k}$

$\vec{v}=\hat{i}-2\hat{j}+3\hat{k}$

So, $\vec{F}=q(\vec{v}\times\vec{B})=q(-5\hat{i}+11\hat{j}+9\hat{k})$

$|F|=q\sqrt{227}$

$R=\frac{|v|^2}{a_C}=2\times10^3m=2$ km

14. 2

Sol. $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$k_{eq}A = k_1A_1 + k_2A_2 + k_3A_3$

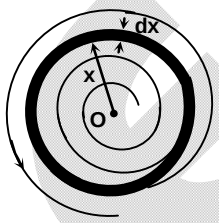
$k_{eq} = \frac{k_1 + k_2 + k_3}{3} = 2$ W/mK

15. 3

Sol. $emf = \frac{d\phi}{dt} = \pi x^2 \frac{dB}{dt}(dN)$

$= \frac{N}{b-a}\pi\left(\frac{dB}{dt}\right)\int_a^b x^2 dx = \frac{14}{3}$ volt

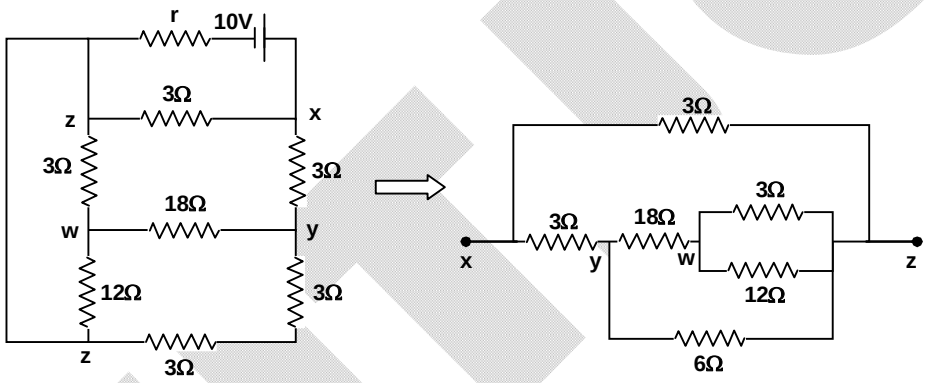
$i = \frac{emf}{R} = 3$ A



16. 7

Sol. R_{eq} between points x and z

$R_{eq} = \frac{28}{13}\Omega$



17. 6

Sol. Immediately after the switching, current in the branch of inductor will not change and charge on capacitor will be zero.

18. 3

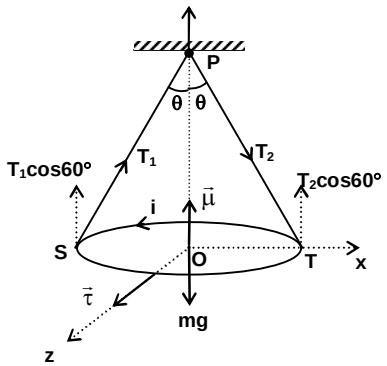
Sol. $T_1\cos60° + T_2\cos60° = mg$

$T_1 + T_2 = 2mG$ …(i)

Taking torque about 'O'

$(T_2\cos60°)R + |\vec{\mu}\times\vec{B}| = (T_1\cos60°)R$ …(ii)

On solving $\frac{T_1}{T_2} = 3$



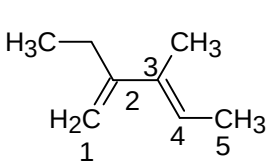
Chemistry

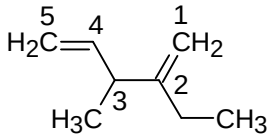
PART – II

Section – A

19. D

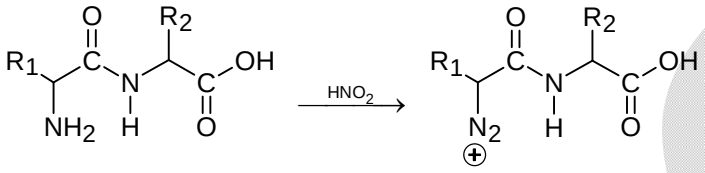
Sol. Option (D) has longest chain of 5C-atom which is same as in the compound given



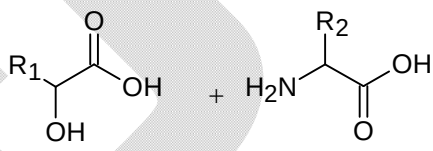


20. C

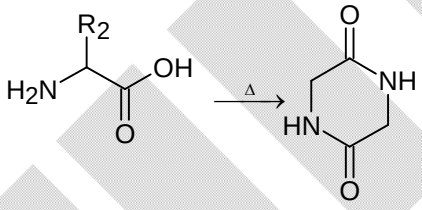
Sol.







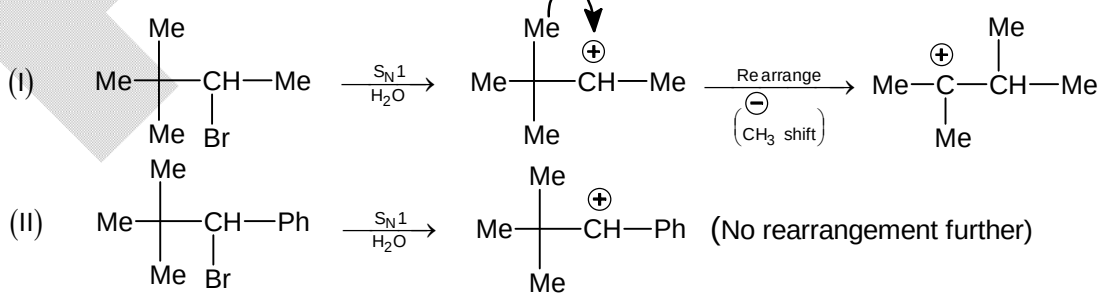
Lactic acid ($R_1 = CH_3$) [y]



2,5-diketopiperazine(R_2 = H)

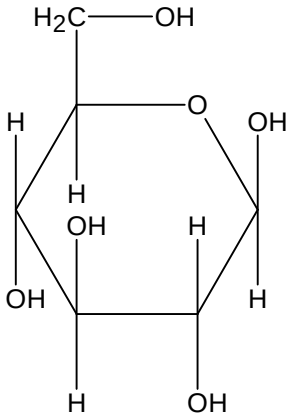
21. B

Sol.



22. C

Sol.

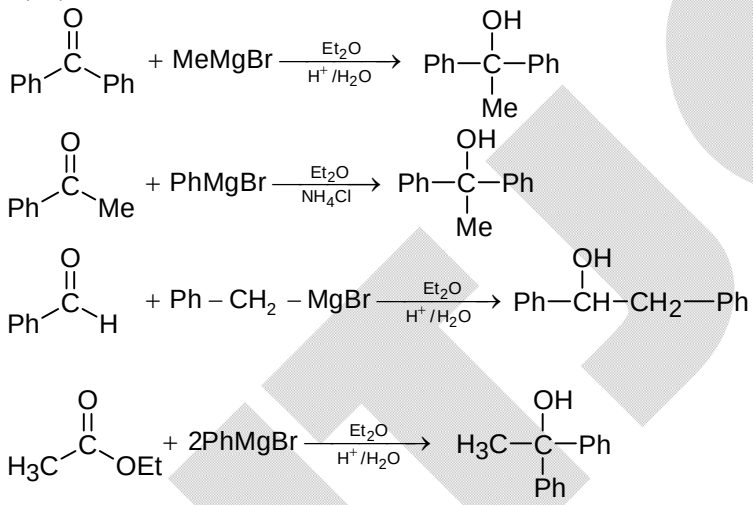


gives positive Tollen's test and NH_2OH/H^+ due to presence of Hemiacetal linkage.

β – D – Glucopyranose

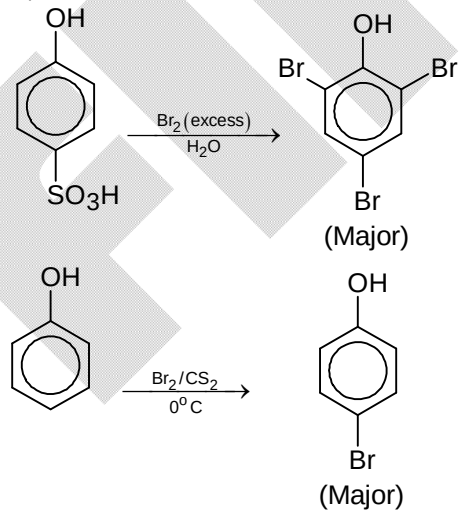
23. A, B, D

Sol.



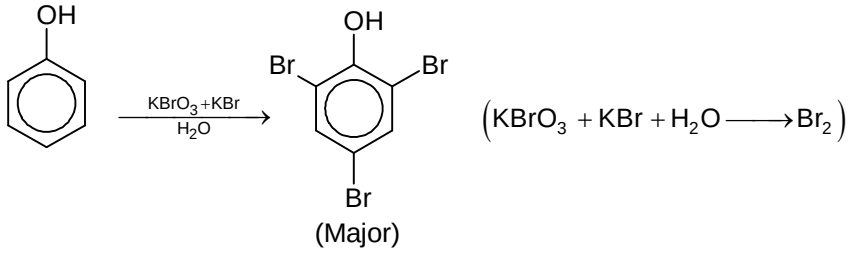
24. A, C

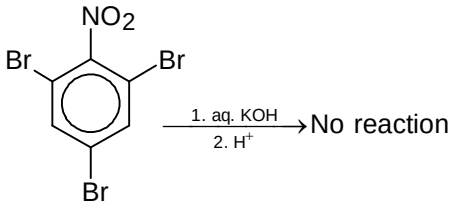
Sol.



(Major)

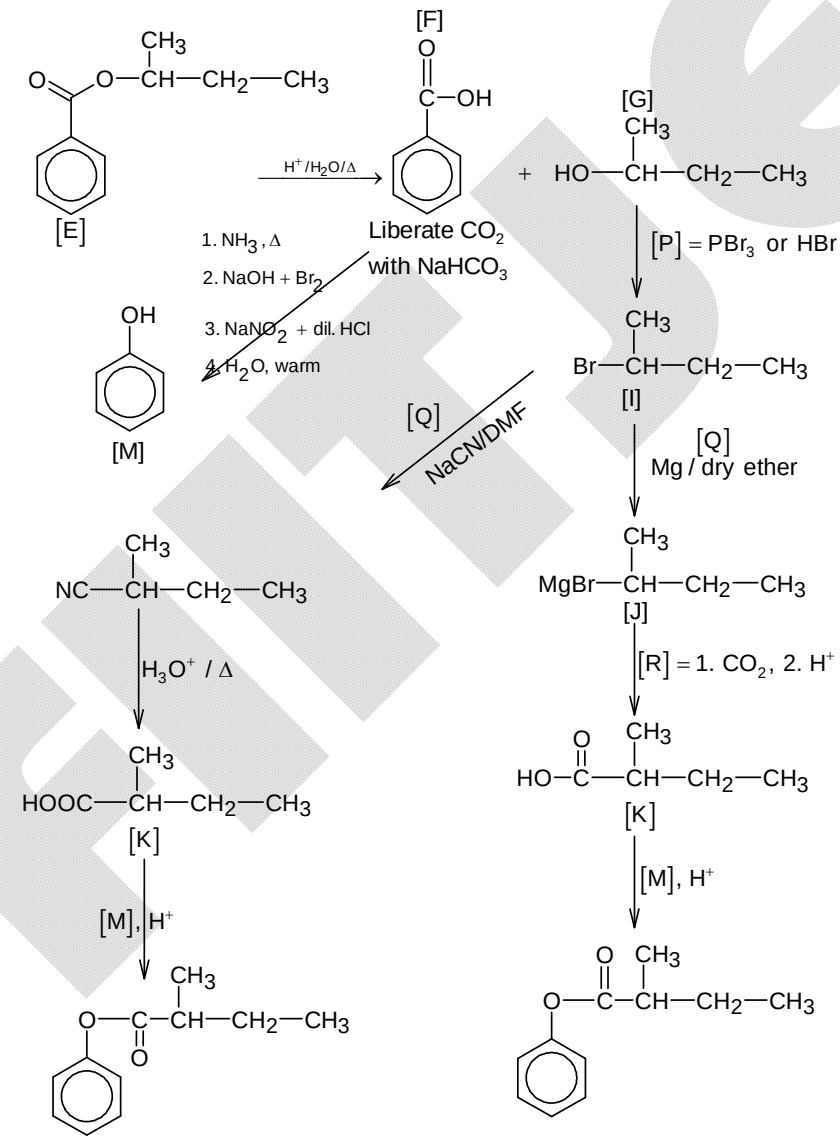
(Major)





25. C, D

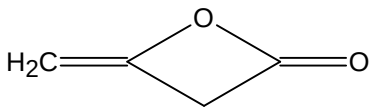
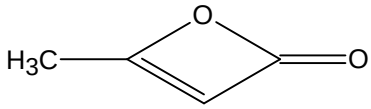
Sol.



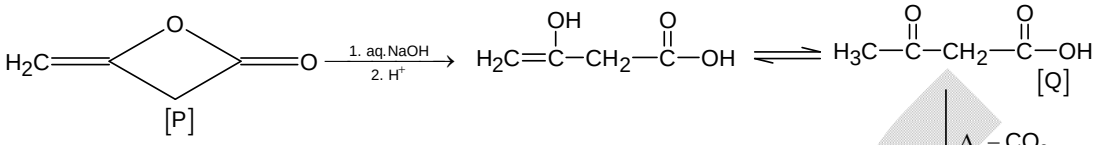
[L] is metamer of [E]

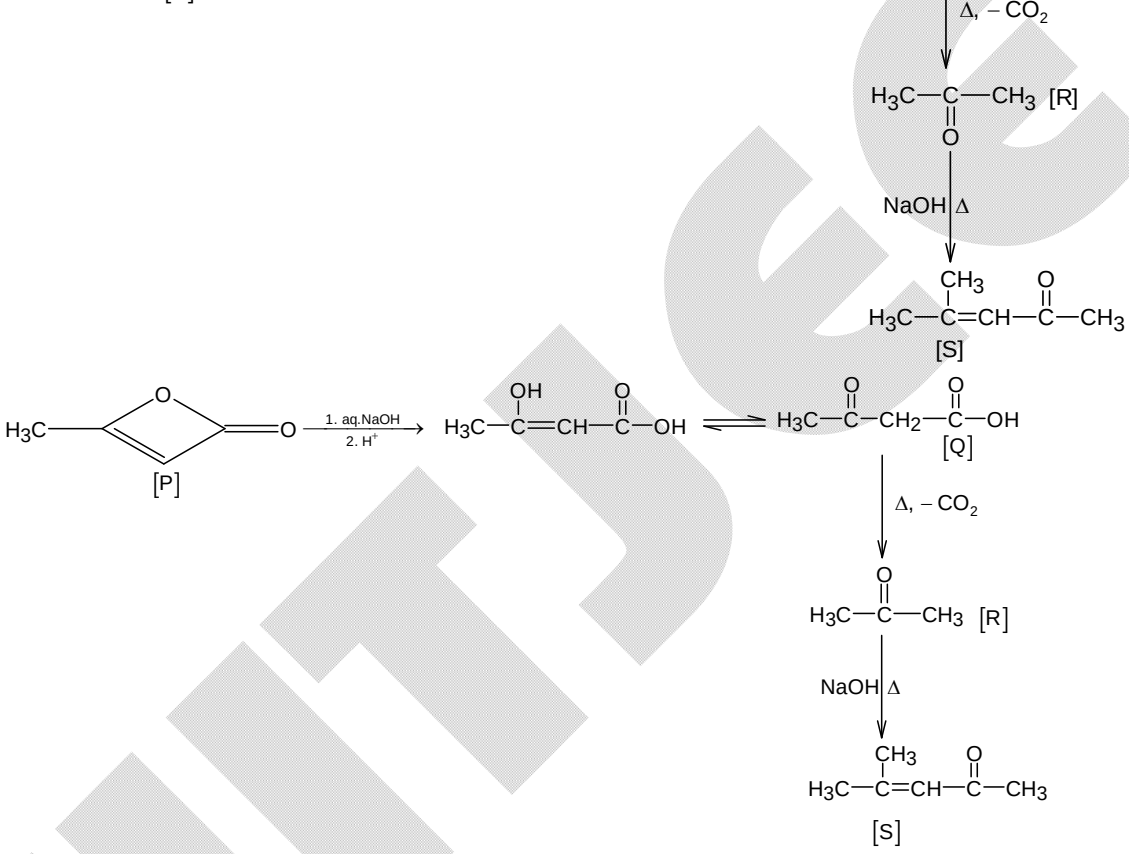
26. A, B, D

Sol.

[P] can be  and 

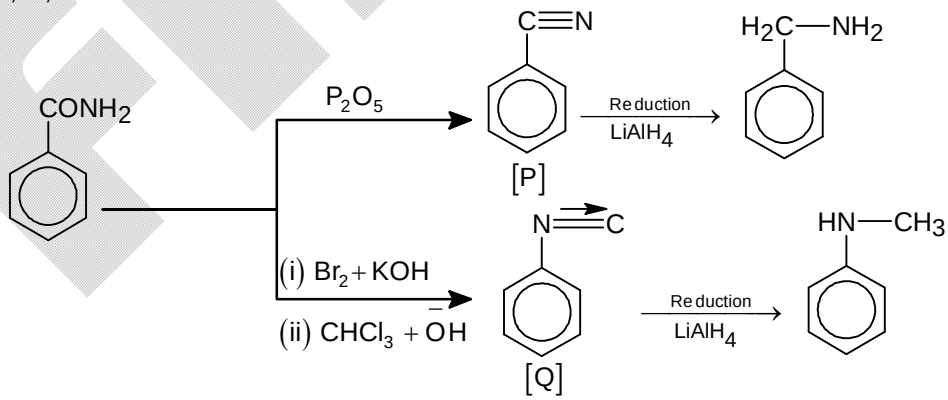
(Two structural isomers)

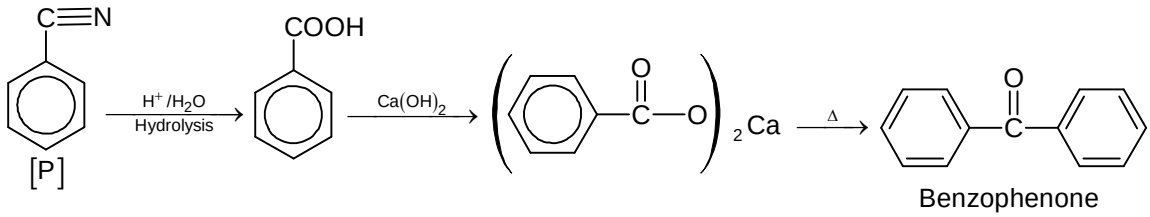


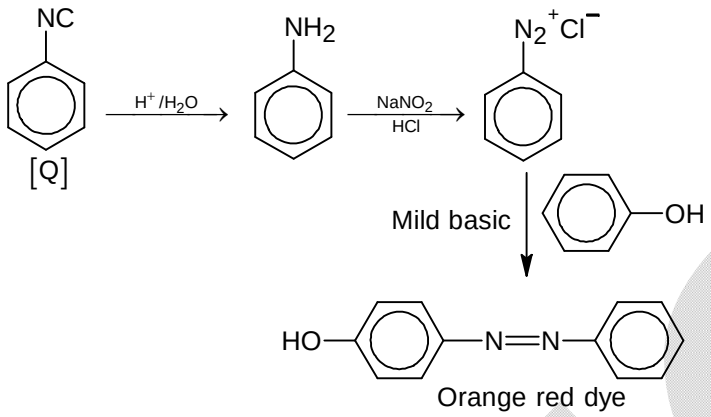


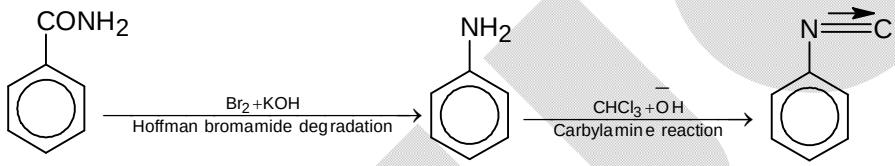
27. B, C, D

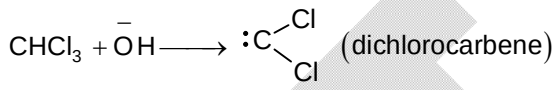
Sol.





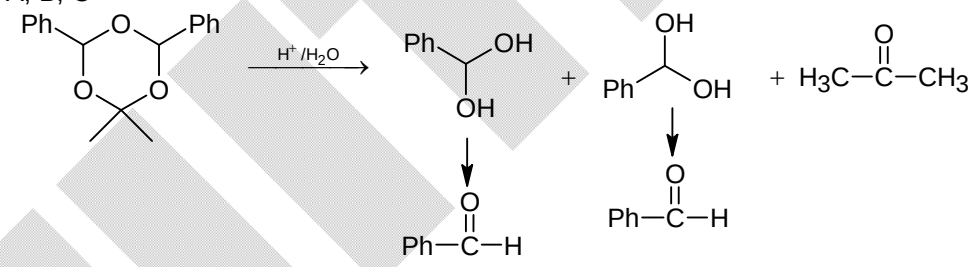


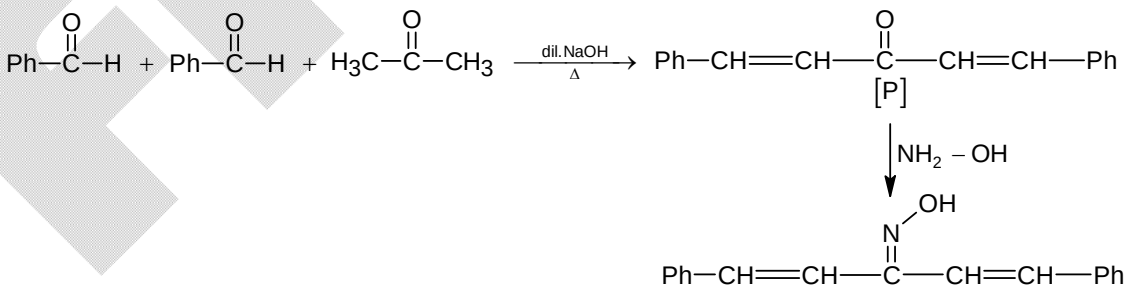




28. A, B, C

Sol.



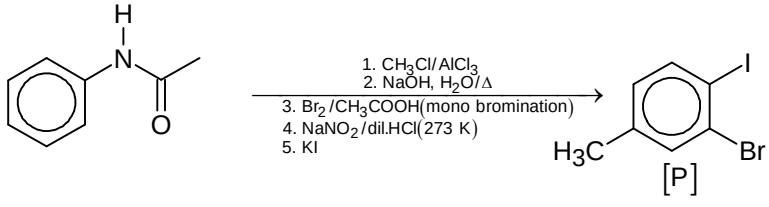


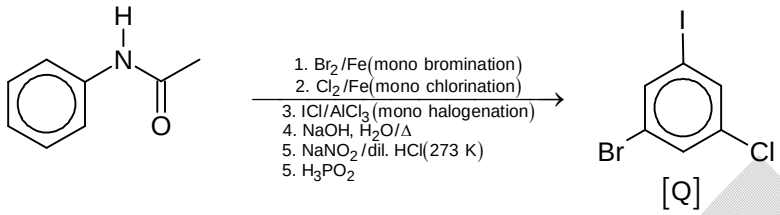
(Number of stereoisomers = 4)

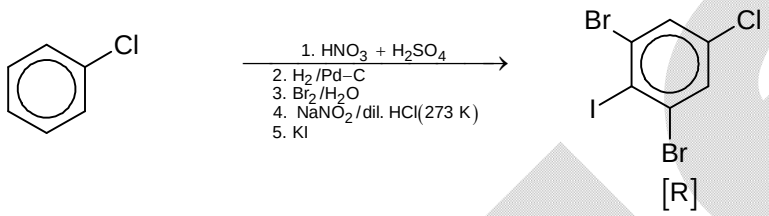
Section – B

29. 9

Sol.





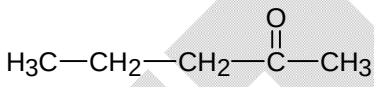


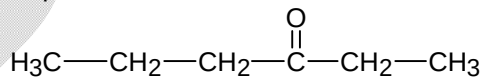
30. 3

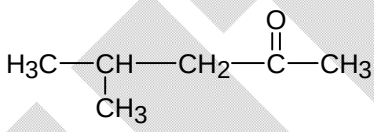
Sol. I, III and VI compounds satisfy the conditions

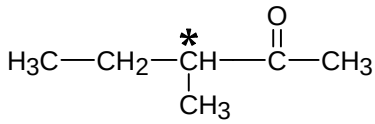
31. 2

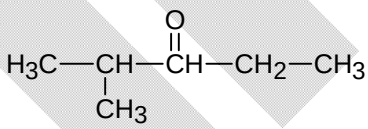
Sol. General formula of isomeric ketone of MW = 100 is $C_6H_{12}O$ possible structures:

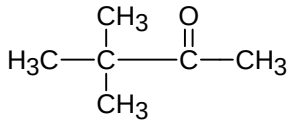
(A) 

(B) 

(C) 

(D) 

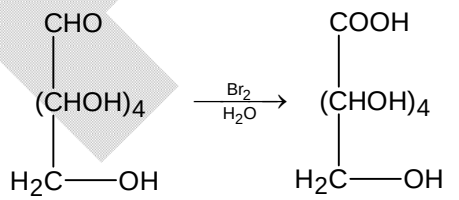
(E) 

(F) 

Compound B and E will give racemic mixture.

32. 4

Sol.



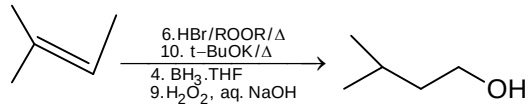
D-Glucose Gluconic acid

33. 7

Sol. Statements 1, 2, 4, 5, 6, 7 and 10 are correct.

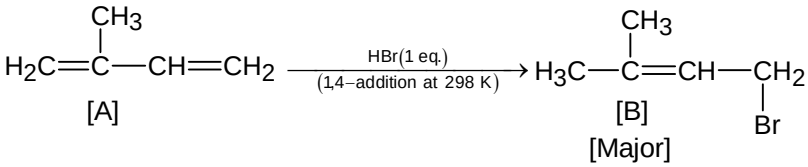
34. 9

Sol.



35. 5

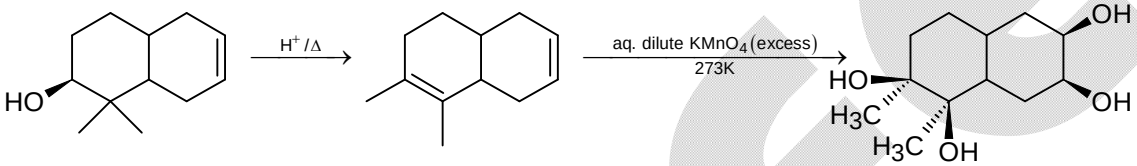
Sol.



[A] [B] [Major]

36. 2

Sol.



[P] [Q]

Mathematics

PART – III

Section – A

37. A

Sol. $$\frac{\frac{\cos A}{\sin A}}{\frac{(\cos B\sin C+\sin B\cos C)}{\sin B\sin C}}=\frac{\cos A\sin B\sin C}{\sin^2 A}=\frac{\left(\frac{b^2+c^2-a^2}{2bc}\right)\frac{b}{2R}\times\frac{c}{2R}}{\frac{a^2}{4R^2}}=\frac{57a^2}{8\times 2bc}\times\frac{bc}{a^2}=\frac{57}{16}$$

38. B

Sol. $DE = AK = \sqrt{(r_1+r_2)^2-(r_2-r_1)^2} = 2\sqrt{r_1r_2}$

Similarly, $BL = EF = 2\sqrt{r_2r_3}$

Let $\angle APD = \theta$ and $PD = x$, then

$$\tan\theta = \frac{AD}{PD}=\frac{BE}{PE}=\frac{CF}{PF}$$

$$\Rightarrow \frac{r_1}{x}=\frac{r_2}{x+2\sqrt{r_1r_2}}=\frac{r_3}{x+2\sqrt{r_1r_2}+2\sqrt{r_2r_3}}=k \text{ (say)}$$

$r_1 = kx,\ r_2 = kx+2k\sqrt{r_1r_2}$, $r_3 = kx+2k\sqrt{r_1r_2}+2k\sqrt{r_2r_3}$; $\frac{r_2-r_1}{2\sqrt{r_1r_2}}=\frac{r_3-r_2}{2\sqrt{r_2r_3}} \Rightarrow r_2=\sqrt{r_1r_3}=\sqrt{4\times 9}=6$

39. C

Sol. $\frac{x^2}{25}+\frac{y^2}{9}=1$ $\therefore$ S is (4, 0)

Let image of S in any tangent be P(h, k),

Midpoint of SP will lie on auxiliary circle

($\because$ foot of perpendicular from focus on any tangent lies on auxiliary circle)

$\therefore \left(\frac{h+4}{2},\frac{k}{2}\right)$ lies on $x^2 + y^2 = 25$ $\therefore$ locus of image is $(x+4)^2 + y^2 = 100$

It touches $ky - 2x + 22 = 0$ $\therefore$ $\frac{|0+8+22|}{\sqrt{k^2+4}}=10 \Rightarrow k^2 + 4 = 9 \Rightarrow k = \pm\sqrt{5}$

40. C

Sol. $\angle AOB = 90°$ (as AB is diameter)

$\therefore$ Slope of OA $\times$ Slope of OB = –1

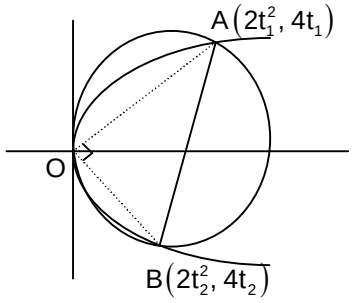
$\frac{2}{t_1}\times\frac{2}{t_2}=-1 \Rightarrow t_1t_2 = -4$

Point of intersection of tangent at A and B is

$C(2t_1t_2, 2(t_1 + t_2)) = C(–8, 2(t_1 + t_2))$

If perpendicular tangents are drawn from C to

$y^2 = kx$, then C must be on its directrix $\therefore$ k = 32



41. A, B, C

Sol. $f(3)=\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}$ and $g(3)=\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{8\pi}{7}$

$\because \cos\frac{6\pi}{7} = \cos\frac{8\pi}{7} \therefore f(3) = g(3)$

$f(6) = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{8\pi}{7} + \cos\frac{10\pi}{7} + \cos\frac{12\pi}{7}$ =

$2\left(\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7}\right) = 2f(3)$

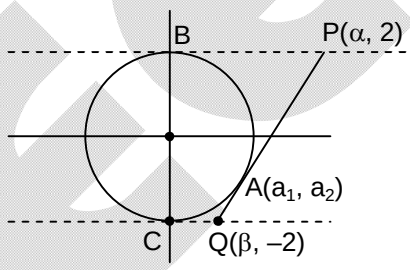
($\because \cos\theta = \cos(2\pi - \theta)$)

$g(6) = \cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{8\pi}{7} + \cos\frac{16\pi}{7} + \cos\frac{32\pi}{7} + \cos\frac{64\pi}{7}$

$\left(\because \cos\frac{64\pi}{7} = \cos\frac{8\pi}{7}, \cos\frac{32\pi}{7} = \cos\frac{4\pi}{7}, \cos\frac{16\pi}{7} = \cos\frac{2\pi}{7}\right) \therefore g(6) = 2g(3)$

42. A, B, C, D

Sol. B(0, 2) and C is (0, –2). Slope of PQ = $\frac{4}{\alpha - \beta}$



Equation of PQ $y - 2 = \frac{4}{\alpha - \beta}(x - \alpha)$

$y(\alpha - \beta) - 2(\alpha - \beta) = 4x - 4\alpha$

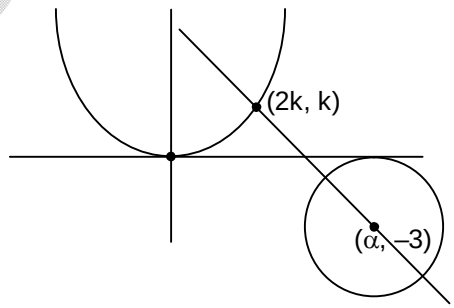
$4x - y(\alpha - \beta) - 2(\alpha + \beta) = 0$

$\because$ it touches $x^2 + y^2 = 4 \therefore \frac{|0 - 0 - 2(\alpha + \beta)|}{\sqrt{16 + (\alpha - \beta)^2}} = 2$

$\Rightarrow \alpha\beta = 4 \Rightarrow$ If $\alpha = 4$ slope of $PQ = \frac{4}{3} \therefore$ Slope of $OA = -\frac{3}{4} \Rightarrow \frac{a_2}{a_1} = -\frac{3}{4}$

43. A, D

Sol. $X^2 = 4ky \Rightarrow 2x = 4k\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2k} \Rightarrow \left(\frac{dy}{dx}\right)_{(2k, k)} = 1$



$\therefore$ Slope of normal at (2k, k) = –1

$\therefore$ Equation of normal $y - k = -1(x - 2k) \Rightarrow x + y = 3k$

If k = 1, equation of normal x + y = 3

$\because$ Shortest distance lies along the common normal

$\therefore$ It passes through $(\alpha, -3)$; $\alpha - 3 = 3 \Rightarrow \alpha = 6$

$\Rightarrow$ If k = 2, x + y = 6 passes through $(\alpha, -3) \therefore \alpha = 9$

$\Rightarrow$ If $\alpha = 20 \therefore x + y = 3k$ passes through (20, –3)

$\therefore k = \frac{17}{3}$

44. B, C

Sol. $s - p = \frac{s - q}{4} = \frac{s - r}{6} = k\,(\text{say}) \because s = \frac{p + q + r}{2} \Rightarrow q + r - p = 2k$; $p + r - q = 8k$; $p + q - r = 12k$

$\Rightarrow p = 10k, r = 5k, q = 7k \therefore S = 11k \because \Delta = \sqrt{s(s-p)(s-q)(s-r)}$

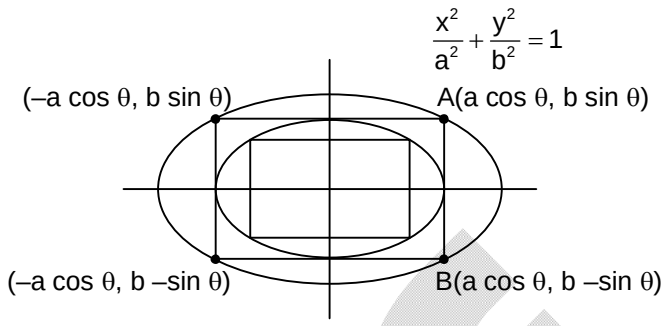
$\Rightarrow \Delta = \sqrt{11k \times k \times 4k \times 6k} = 2\sqrt{66}\,k^2$; Circumradius = $\frac{pqr}{4\Delta} \Rightarrow \frac{175}{4\sqrt{66}} = \frac{5k \times 7k \times 10k}{4 \times 2\sqrt{66}\,k^2} \Rightarrow k = 1$

$\therefore$ Area = $2\sqrt{66}$ inradius = 4(Circumradius) $\sin\frac{P}{2}\sin\frac{Q}{2}\sin\frac{R}{2}$

$\Rightarrow 2\sqrt{\frac{6}{11}} = 4 \times \frac{175}{4\sqrt{66}} \times \sin\frac{P}{2}\sin\frac{Q}{2}\sin\frac{R}{2}$

45. A, C, D

Sol. Area of inscribed rectangle
$= 2a\cos\theta \times 2a\sin\theta = 2ab\sin 2\theta$
(Here, $a = 4\sqrt{2}; b = 2\sqrt{2}$)
Maximum value = 2ab when $\theta = 45°$
$\therefore$ Area of R_1 = 2ab
and sides of r_1 are $a\sqrt{2}; b\sqrt{2}$
Let major axis and minor axis of E_2 be $2a_2$, $2b_2$



$\therefore\ 2a_2 = a\sqrt{2} \Rightarrow a_2 = \frac{a}{\sqrt{2}}; b_2 = \frac{b}{\sqrt{2}}$

$\therefore$ Area of $R_2 = 2\times\frac{a}{\sqrt{2}}\times\frac{b}{\sqrt{2}} = ab$ and sides are of R_2 are a and b

$\vdots$

Similarly area of $R_n = \frac{2ab}{2^{n-1}}$ and sides of R_n are $\frac{\sqrt{2}a}{(\sqrt{2})^{n-1}}; \frac{\sqrt{2}b}{(\sqrt{2})^{n-1}}$

Similarly major axis and minor axis of E_n are $\frac{2a}{(\sqrt{2})^{n-1}}; \frac{2b}{(\sqrt{2})^{n-1}}$

$\therefore$ Sum of areas of rectangles R_1, R_2, R_3 ..... is $2ab + ab + \frac{ab}{2}+\frac{ab}{4}+.....= \frac{2ab}{1-\frac{1}{2}} = 4ab$

$\therefore$ Sum of areas is always less than $4\times 4\sqrt{2}\times 2\sqrt{2} = 64$
Eccentricities does not change $\therefore\ E_{10} = E_{12}$

46. A, C, D

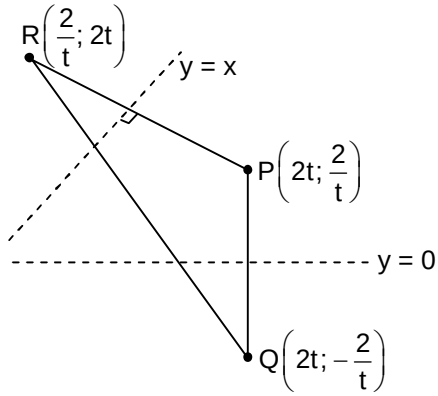
Sol. Centroid : $\left(\frac{4t+\frac{2}{t}}{3}; \frac{2t}{3}\right)$; $h = \frac{4t+\frac{2}{t}}{3}; k = \frac{2t}{3}$

$\Rightarrow t = \frac{3k}{2}\ \therefore\ 3h = 6k + \frac{4}{3k} \Rightarrow 9hk = 18k^2 + 4$

Circumcentre is (0, 0)

$\therefore$ Orthocentre : $\left(4t+\frac{2}{t}; 2t\right)$

$\therefore$ Locus $hk = 2k^2 + 4$



Section – B

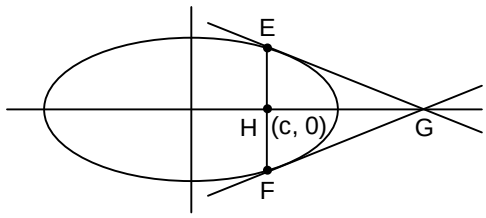
47. 2

Sol. $T_r = \frac{1}{3^{r-1}}\sin^3(3^{r-1}\theta) = \frac{1}{4\times 3^{r-1}}(3\sin 3^{r-1}\theta - \sin 3^r\theta) = \frac{1}{4}\left(\frac{\sin 3^{r-1}\theta}{3^{r-2}} - \frac{\sin 3^r\theta}{3^{r-1}}\right) = \frac{1}{4}(V_r - V_{r+1})$

$\therefore\ S_{10} = \frac{1}{4}(V_1 - V_2 + V_2 - V_3 + ... V_{10} - V_{11}) = \frac{1}{4}(V_1 - V_{11}) \Rightarrow f(\theta) = \frac{1}{4}\left(3\sin\theta - \frac{\sin 3^{10}\theta}{3^9}\right)$

48. 8

Sol. Let E be $(5\cos\theta, 3\sin\theta) \Rightarrow 5\cos\theta$



$\Rightarrow c \Rightarrow \cos\theta = \frac{c}{5}$ F is $(5\cos\theta, -3\sin\theta)$

Tangent at E : $\frac{x\cos\theta}{5} + \frac{y\sin\theta}{3} = 1$

Tangent at F : $\frac{x\cos\theta}{5} - \frac{y\sin\theta}{3} = 1 \ \therefore$ G is $\left(\frac{5}{\cos\theta}, 0\right)$

Area of $\Delta EFG = \frac{1}{2}\times GH\times EF = \frac{1}{2}\times\left(\frac{5}{\cos\theta} - 5\cos\theta\right)\times 6\sin\theta$

$= \frac{1}{2}\times\left(\frac{25}{c} - c\right)\times 6\left(\sqrt{1-\frac{c^2}{25}} = \frac{3}{5c}(25-c^2)^{3/2}\right) \Rightarrow \Delta'(c) = \frac{3}{5}\left(\frac{c\frac{3}{2}(25-c^2)^{1/2}\times(-2c) - (25-c^2)^{3/2}}{c^2}\right)$

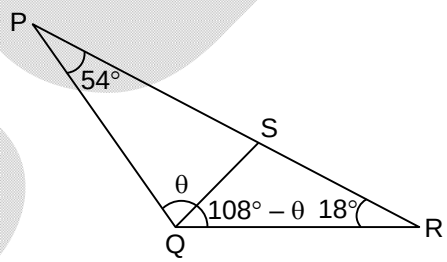
Which is always negative (–ve) $\therefore \Delta(c)$ is always decreasing

If $1 \le c \le 3$, then minimum value of $\Delta(c)$ is at $c = 3$

$\Delta(3) = \frac{3}{5\times 3}(16)^{3/2} = \frac{3}{5\times 3}\times 2^6 = \frac{64}{5}$

49. 5

Sol. $\because$ Area of $\Delta PSQ = (\sqrt{5}-1)$ area of ΔRSQ



$\frac{1}{2}\times PQ\times QS\times\sin\theta = (\sqrt{5}-1)\times\frac{1}{2}QR\times QS\times\sin(108-\theta)$

$\because \frac{PQ}{\sin 18^\circ} = \frac{QR}{\sin 54^\circ}$

$\therefore \sin\theta\sin 18^\circ = (\sqrt{5}-1)\sin 54^\circ\sin(108^\circ-\theta)$

$\therefore \sin\theta\frac{\sqrt{5}-1}{4} = (\sqrt{5}-1)\times\frac{\sqrt{5}+1}{4}\times\sin(108^\circ-\theta) \Rightarrow \frac{\sin(108^\circ-\theta)}{\sin\theta} = \frac{1}{\sqrt{5}+1} = \frac{\sqrt{5}-1}{4}$

$\sin 108^\circ\cot\theta - \cos 108^\circ = \frac{\sqrt{5}-1}{4} \Rightarrow \cot\theta = 0 \Rightarrow \theta = 90^\circ$

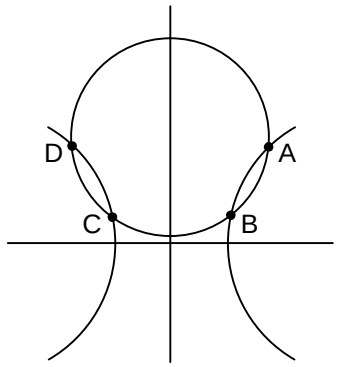
50. 8

Sol. Let A be $(12\sec\theta_1, 12\tan\theta_1)$

B be $(12\sec\theta_2, 12\tan\theta_2)$

C be $(12\sec\theta_3, 12\tan\theta_3)$ and

D be $(12\sec\theta_4, 12\tan\theta_4)$



Let centroid of ABC be (h, k)

$\Rightarrow h = 4\sec\theta_1 + 4\sec\theta_2 + 4\sec\theta_3$ and

$K = 4\tan\theta_1 + 4\tan\theta_2 + 4\tan\theta_3$

$\frac{12\sec\theta_1 + 12\sec\theta_2 + 12\sec\theta_3 + 12\sec\theta_4}{4} = \frac{6+0}{2}$

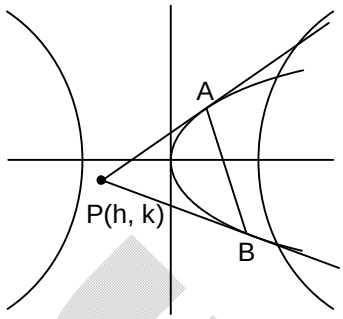
$3h + 12\sec\theta_4 = 12 \Rightarrow 4\sec\theta_4 = 4 - h$

$\frac{12\tan\theta_1 + 12\tan\theta_2 + 12\tan\theta_3 + 12\tan\theta_4}{4} = \frac{12+0}{2}$

$3k + 12\tan\theta_4 = 24 \Rightarrow 4\tan\theta_4 = 8 - k \Rightarrow (h-4)^2 - (k-8)^2 = 16 \Rightarrow \alpha = 4, \beta = 8, \gamma = 4$

51. 3

Sol. Equation of chord of contact AB $ky = 2(x + h)$



$y = \frac{2x}{k} + \frac{2h}{k}$ $\because$ it touches $\frac{x^2}{9} - \frac{y^2}{16} = 1$

$\frac{4h^2}{k^2} = \frac{9 \times 4}{k^2} - 16$; $h^2 + 4k^2 = 9$

$\Rightarrow$ locus is $\frac{x^2}{9} + \frac{y^2}{9/4} = 1$

$\frac{9}{4} = 9(1 - e^2) \Rightarrow e = \frac{\sqrt{3}}{2}$

52. 2

Sol. $\because$ It is equilateral Δ $\therefore$ $x_1 = \frac{a}{3}(\cos\alpha + \cos\beta + \cos\gamma)$; $y_1 = \frac{b}{3}(\sin\alpha + \sin\beta + \sin\gamma)$

$\frac{9x_1^2}{a^2} + \frac{9y_1^2}{b^2} = (\cos\alpha + \cos\beta + \cos\gamma)^2 + (\sin\alpha + \sin\beta + \sin\gamma)^2$

$= 3 + 2(\cos\alpha\cos\beta + \cos\beta\cos\gamma + \cos\gamma\cos\alpha + \sin\alpha\sin\beta + \sin\beta\sin\alpha + \sin\gamma\sin\alpha)$

53. 6

Sol. $\therefore$ $C_1 : x^2 + y^2 = a^2 - b^2 \Rightarrow C_2 : x^2 + y^2 = \frac{a^2 + b^2}{3}$

Director circle of C_1 is $x^2 + y^2 = 2(a^2 - b^2) \Rightarrow 2(a^2 - b^2) = \frac{a^2 + b^2}{3}$

$\Rightarrow 6a^2 - 6b^2 = a^2 + b^2 \Rightarrow 5a^2 = 7b^2$

$5a^2 = 7a^2(1 - e_1^2) \Rightarrow e_1^2 = \frac{2}{7}$ and $5a^2 = 7a^2(e_2^2 - 1) \Rightarrow e_2^2 = \frac{12}{7}$

54. 7

Sol. $(a - 2b)^2 + (a - 3c)^2 = 0 \Rightarrow a = 2b, a = 3c \Rightarrow b = \frac{a}{2}$, $c = \frac{a}{3}$

$\cos B = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow \cos B = \frac{a^2 + \frac{a^2}{9} - \frac{a^2}{4}}{2 \times a \times \frac{a}{3}} = \frac{31}{24}$